

Finding Diagonal Measures in Parallelograms

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A **parallelogram** is a quadrilateral with both pairs of opposite sides parallel.

Diagonals

The **diagonals** of a parallelogram intersect at a point called the **midpoint** of each diagonal. The diagonals of a parallelogram **bisect each other**.

Bisecting Diagonals

If diagonals AC and BD intersect at point E ,
then:

$$AE = EC \quad \text{and} \quad BE = ED$$

The two halves of each diagonal are congruent.

Finding Missing Measures

To find missing measures, set the two halves of a diagonal equal to each other and solve the resulting equation.

Finding an Entire Diagonal

Once one half of a diagonal is known, the entire diagonal can be found by adding the two congruent halves.

$$AC = AE + EC$$

$$BD = BE + ED$$

Examples

Example 1

In parallelogram $ABCD$, diagonals AC and BD intersect at E . If $AE = 8$, find EC .

Since the diagonals bisect each other,

$$AE = EC$$

Therefore,

$$EC = 8$$

Example 2

In parallelogram $ABCD$, diagonals AC and BD intersect at E . If $AE = 3x + 1$ and $EC = x + 11$, find x .

$$3x + 1 = x + 11$$

$$2x = 10$$

$$x = 5$$

Example 3

In parallelogram $ABCD$, diagonals AC and BD intersect at E . If $AE = 9$, find the length of diagonal AC .

Since the diagonals bisect each other,

$$EC = 9$$

Therefore,

$$AC = 9 + 9 = 18$$

Example 4

In parallelogram $PQRS$, diagonals intersect at T . If $PT = 4x - 1$ and the entire diagonal $PR = 30$, find x .

Since the diagonals bisect each other,

$$PT = \frac{PR}{2} = 15$$

Thus,

$$4x - 1 = 15$$

$$4x = 16$$

$$x = 4$$

Practice Exercises

Practice 1

In parallelogram $ABCD$, diagonals AC and BD intersect at E . If $AE = 8$, find EC .

Practice 2

In parallelogram $WXYZ$, diagonals intersect at M . If $WM = 13$, find MY .

Practice 3

In parallelogram $PQRS$, diagonals intersect at T . If $BT = 11$, and B and D are endpoints of one diagonal, find TD .

Practice 4

In parallelogram $ABCD$, diagonals intersect at E . If $AE = 3x + 1$ and $EC = x + 11$, find x .

Practice 5

In parallelogram $ABCD$, diagonals intersect at E . If $AE = 9$, find the length of diagonal AC .

Practice 6

In parallelogram $PQRS$, diagonals intersect at T . If $PT = 4x - 1$ and the entire diagonal $PR = 30$, find x .

Activities

Activity 1

In parallelogram $QRST$, diagonals intersect at U . If $QU = 17$, find US .

Activity 2

In parallelogram $WXYZ$, diagonals intersect at K . If $WK = x + 9$ and $KY = 3x - 1$, find x .

Activity 3

In parallelogram $PQRS$, diagonals intersect at T . If $PT = 4x - 3$ and $TR = 2x + 9$, find x .

Activity 4

In parallelogram $JKLM$, diagonals intersect at N . If $JN = 6x - 8$ and $NL = 4x + 6$, find x .

Activity 5

In parallelogram $QRST$, diagonals intersect at U . If $BU = 18$, where B and D are endpoints of one diagonal, find the length of diagonal BD .

Activity 6

In parallelogram $PQRS$, diagonals intersect at T . If $QT = 2x + 10$ and $TS = 4x - 8$, find x and then find the length of diagonal QS .

Thank you!